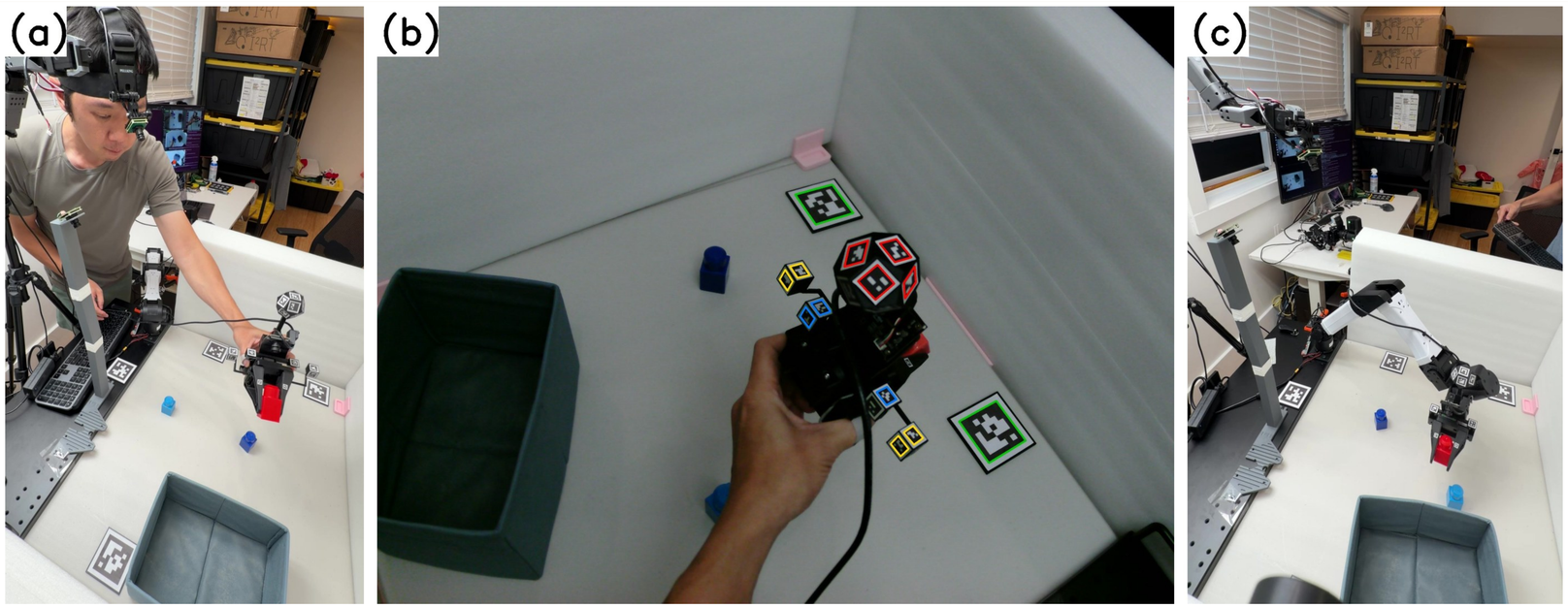
A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

Yosub Shin[1] Yuan Liu[2] Igor Molybog[1] [1]University of Hawai'i at Mānoa [2]Independent

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One camera, two mounts



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 arm beside the robot.

Why the head

- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

Labels from one frame

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point — **1.6°** orientation — **3.5 %** jaw width, of stroke

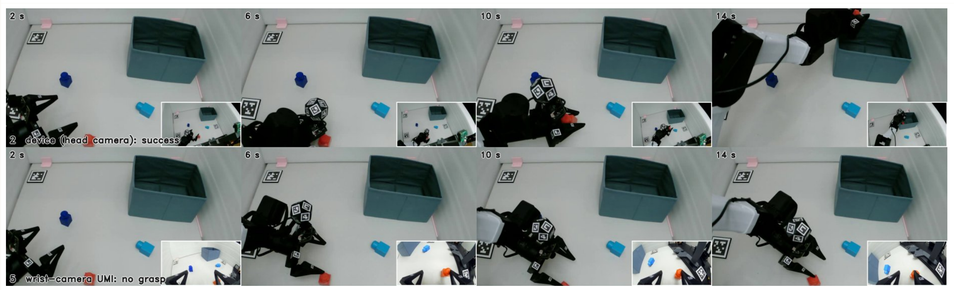
Median error against the robot's encoders. A label on 93–100 % of frames.

The robot's head

- A $122 six-joint arm carries the same camera beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts.

46 mm head pose vs the collector — **5.4°** static camera: 87 mm, 7.0°

One placement, two views



Scene 16 at 2, 6, 10 and 14 s, each policy's own view inset. Top: the head-camera policy keeps gripper and bin in one frame. Bottom: the wrist policy sees the block and the wall, and drops it.

Collection

$144 glove, camera, strap — **103 g** on the head

2–3× demos per minute vs teleop — **198** demos in four sessions

Eight policies, one recipe

ACT, joint-space actions, same 194 glove demos in every mix. 25 placements, one trial each.

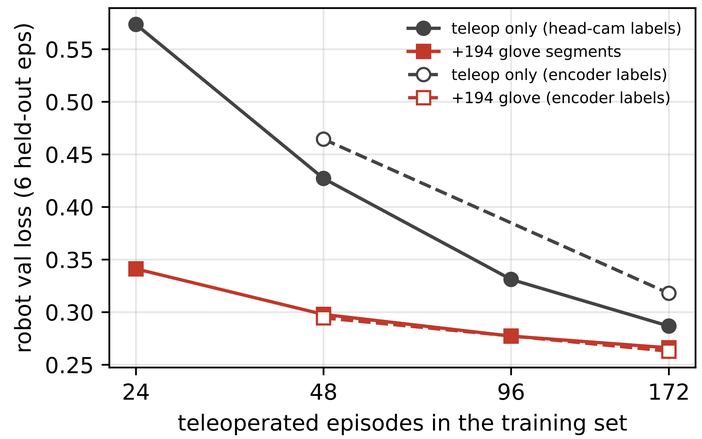
#	Policy (glove labels → observation)	Val	Success
1	teleop only, 172 ep. → head cam	0.318	18/25
2	**device**: head → head, 172 + 194	**0.263**	**20/25**
3	teleop only, 48 ep. → head cam	0.464	10/25
4	**device, low budget**, 48 + 194	0.295	**17/25**
5	wrist UMI: fixed → wrist fisheye	0.257	15/25
6	head-cam labels for encoders	0.328	15/25
7	fixed camera as the system	0.256	12/25
8	head labels → wrist fisheye	0.252	14/25

The view decides how a policy fails

20/25 head camera — **15/25** wrist fisheye (UMI) — **12/25** fixed camera

- Wrist policies **drop the block mid-carry** (12 of 21 failures), always while looking away from the bin. Head policies: 1 of 45.
- Same labels, other view (arms 8 vs 2): 20/25 falls to 14/25. Same view, other labels (8 vs 5): no difference.
- **Head-camera labels** cost 3 % of validation loss against encoders, nothing on the robot (6 vs 1).

194 glove demos ≈ 65–100 teleoperated episodes



Robot validation loss against teleoperation budget, with (dashed) and without the glove set. Added to 48 teleoperated episodes: 10/25 to 17/25, level with 172 alone.

Limits and next

One task, one collector, one tile layout. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.